

Yunxi Jiang

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Professional Experience

Centre National de la Recherche Scientifique(CNRS)
Postdoc Fellow

Jan 2026– Now

Education

University of Liverpool

Sep 2017 – Jun 2021

B.Sc. in Applied Mathematics

University of Liverpool

Dec 2021 – Dec 2025

Ph.D. in Physics

Research Areas

Condensed Matter Physics, Quantum Computing, AI for Physics

Publications

1. **Y. Jiang**, X. Chen, C. Xuan, Z. Wang, H. Yu, Altermagnet Skyrmions: Macroscopic Realization of d -wave Symmetry, *Physical Review B* **112**(1), 014423 (2025).
2. **Y. Jiang**, X. Chen, C. Xuan, H. R. Sharma, H. Yu, Dynamics of current-driven spin waves in a confined track, *Journal of Magnetism and Magnetic Materials* **615**, 172800 (2025).
3. **Y. Jiang**, H. Yu, X. Chen, Skyrmion-based racetrack multilevel data storage device manipulated by pinning, *Journal of Applied Physics* **134**, 053901 (2023).
4. **Y. Jiang**, C. Xuan, H. Yu, Current-driven dynamics of skyrmions in the presence of pinning at finite temperatures, *Journal of Magnetism and Magnetic Materials* **562**, 169786 (2022).
5. **Y. Jiang**, H. Yu, Research progress on dynamics of skyrmions under the effect of pinning, *Science and Engineering* **1**(1), 38–46 (2022).
6. Q. Zhu, **Y. Jiang**, L. Fan, ClassWise-CRF: Category-Specific Fusion for Enhanced Semantic Segmentation of Remote Sensing Imagery, *Neural Networks*, 197 (2026)
7. Q. Zhu, **Y. Jiang**, L. Fan, SO3UFormer: Learning Intrinsic Spherical Features for Rotation-Robust Panoramic Segmentation, (Under review)
8. **Y. Jiang**, C. Xuan, X. Chen, Z. Wang, Q. Zhu, H. Yu, Intrinsic Double-Well Potential of Altermagnetic Skyrmions: Toward a Qubit, (Under review)

Conferences

1. *International Workshop on the Frontiers of Racetrack Memory*, Max Planck Institute, Berlin, German
July 2025
Poster Presentation: Skyrmion-based racetrack memory with altermagnetic and ferromagnetic textures
2. *6th International Symposium on Quantum Physics and Quantum Information Science*, Beijing, China
October 2024
Poster Presentation: *Interplay between skyrmion and magnon*
3. *2022 Chinese Physical Society Fall Meeting*, Shenzhen, China
November 2022
Poster Presentation: *Current-driven dynamics of skyrmions* (Best Poster Award)

Honors and Awards

1. Best Poster Award (Top 5%), *Journal of Magnetism and Magnetic Materials* & 2022 Chinese Physical Society Fall Meeting
2. XJTLU Postgraduate Research Scholarship (PGRS2106030), Xi'an Jiaotong-Liverpool University

Patent

H. Xu, B. Liu, H. Yu, Y. Wang, X. Wang, **Y. Jiang**, *Simulation neuron device and simulation neural network based on skyrmion*, CN115564034 (A), Issued January 3, 2023

Experimental Experience

University of Liverpool

March 2024 – August 2024

- Collaborated with Dr. Hem Raj Sharma and Dr. Liam O'Brien to fabricate multilayer thin films using e-beam evaporation techniques.
- Characterized magnetic anisotropy of thin films via Magneto-Optic Kerr Effect (MOKE) measurements, analyzing the influence of film structure on magnetic properties.

Teaching Experience

Teaching Assistant

- Physics
- Engineering Mathematics
- Statistical Distribution Theory
- Advanced Linear Algebra

Skills

1. *Programming*: C, Python, Java, MATLAB, R
2. *Languages*: English (Fluent), Mandarin Chinese (Native), French (Basic), Spanish (Basic), Cantonese (Basic)
3. *Research Skills*: Micromagnetic Simulations, Density Functional Theory Calculations (VASP, Quantum ESPRESSO), Machine Learning
4. *Experimental Techniques*: E-beam evaporation, Magneto-Optic Kerr Effect (MOKE) measurements